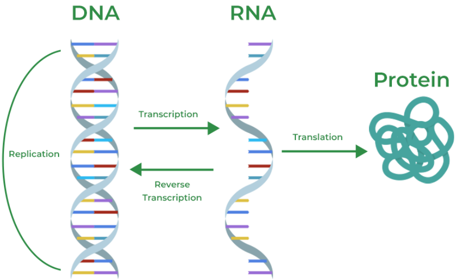
Exercise Design Patterns I

DNA to Protein



Use the following Python code: `dna2protein.py`

1. Translate the given DNA sequence into a protein sequence by using the given code (`transcribe_dna_to_rna` and `translate_rna_to_protein`).

Exercise Design Patterns II

2. Create a utility class, which contains both methods as static methods.
3. Store the sequence into an object of the class SequenceStorage.
4. Change the class SequenceStorage in a way that only one object could exist. Which design pattern could be used?
5. Take the coding sequence of the gene CD28. What is the corresponding protein sequence?
6. Now there is a new requirement. Your program should also work with protein sequences. Extend the code to make that possible.
7. Create a class `SequenceFactory`, which contains a method to create a sequence. Based on a given flag, the sequence will be a DNA or protein sequence.